

Representing Electrophysiological Activity Data using the Neo Library

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Why Data Representations?

Data:

A number

Representation:

1 (a float number)

But that's boring data!

Why Data Representations?

Data:

Many numbers

Representation:

3,4,5 (an array of float)

But that could be anything!

Why Data Representations?

Data:

A number of ice cream scoops

Representation:

5 (an integer)

In Python:

```
ice_cream_scoops = 5
```

But that's an implicit counting unit of “scoops”!

Why Data Representations?

Data:

A temperature

Representation:

38.6 C (a float and a unit)

In Python:

`temperature_in_C = 38.6` *or*
`temperature = {x: 38.6, u: "C"}`

But there are more data!

Why Data Representations?

Data:

A temperature time series

Representation:

[38.6, 32.4] C; [8 pm, 10 pm]
(two arrays and two units)

In Python:

```
times_h = [8, 10]
```

```
temperature_C = [38.6, 32.4]
```

But that's just a part of the experiment!

Why Data Representations?

Data:

Ice cream scoops and
temperature sold as a function
of time

Representation:

Related arrays

In Python:

```
times_temp_h = [8, 10]
```

```
temperature_C = [38.6, 32.4]
```

```
time_H = [20, 21, 22]
```

```
cumul_scoops = [30, 128, 303]
```

But I don't know anything about the measurement!

Why Data Representations?

Data:

Ice cream scoops and temperature sold as a function of time

Representation:

Related arrays + notes

In Python:

```
times_temp_h = [8, 10]
temperature_C = [38.6, 32.4]
time_H = [20, 21, 22]
cumul_scoops = [30, 128, 303]
notes_scoop_sales =
    {"rainy", "price lowered",
     "rich person passing by"}
```


Why Standardized Data Representations?

- **Reducing errors:** Ad-hoc models lack consistency checks, making complex analyses error-prone.
- **Tool/Script interoperability:** Shared models allow scripts to be reused without costly variable conversions.
- **Abstraction of heterogeneous file formats:** File formats are often developed with the data provider in mind – not the data user.
- **Simplifying collaboration:** A common model makes datasets and code easier for collaborators to understand.

Describing Data in Electrophysiology

Required:

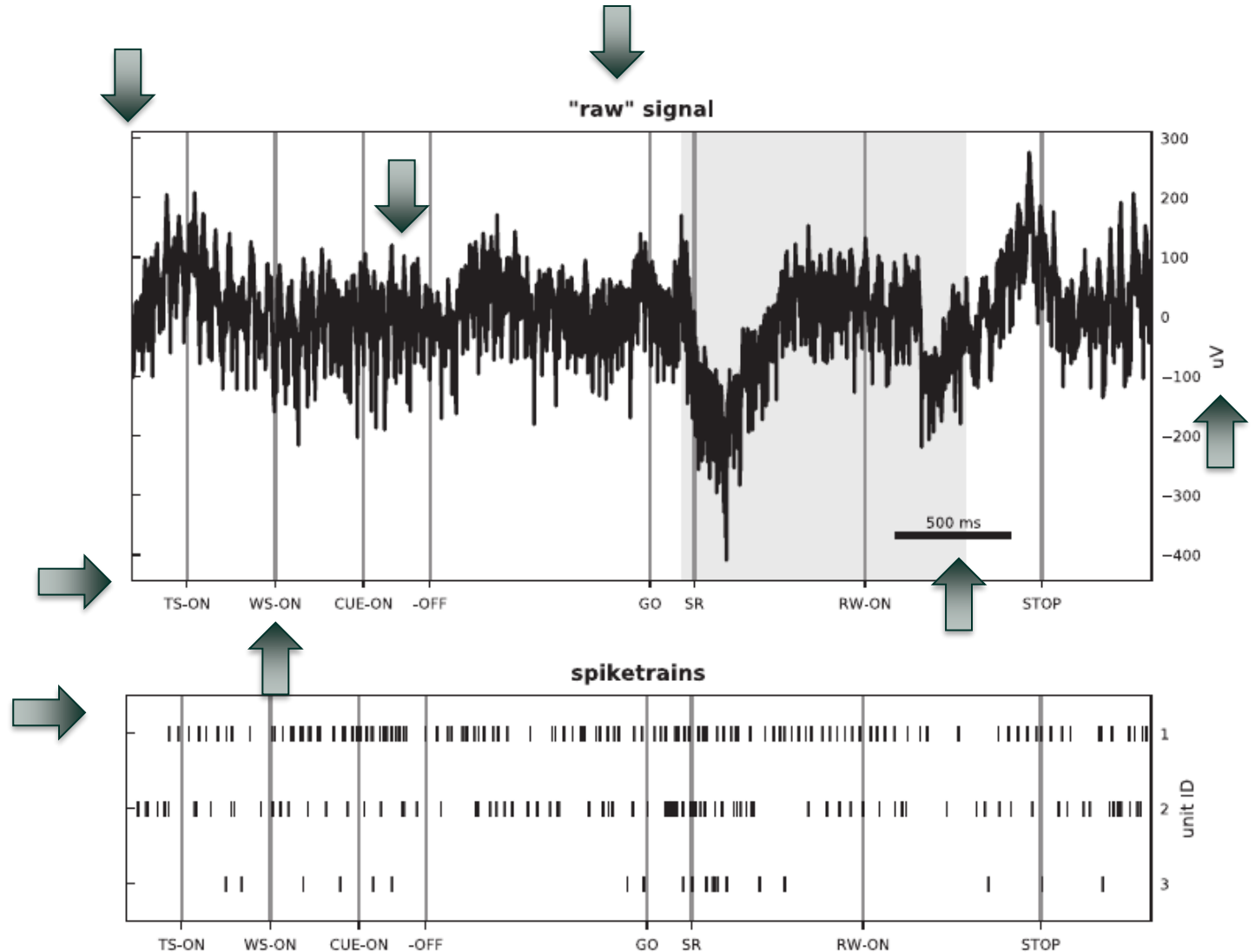
- signal
- time points – *or* – recording start time and sampling rate
- units of measurement

Relationships:

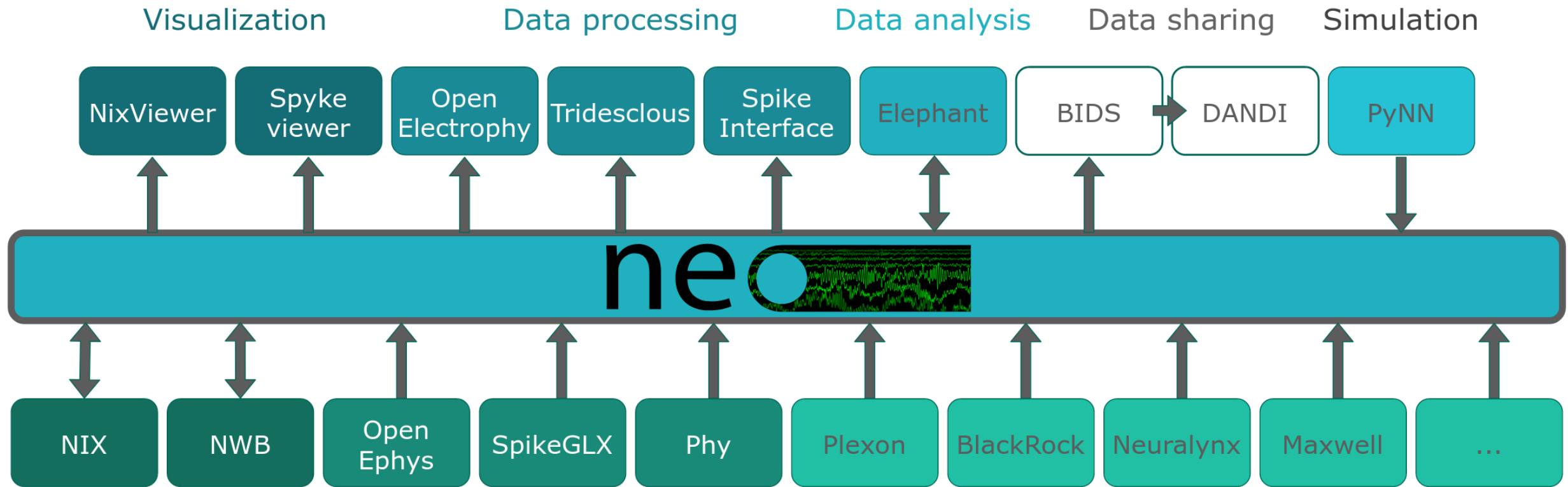
- What's recorded in parallel?
- What signals are related?

Semantics:

- name
- description
- origin
- annotations



Neo as a Bridge between Data and Tools



The Neo Data Model Overview

Represent Data

Relate Data

AnalogSignal

data at regular sampling intervals

SpikeTrain

neuronal time point series

IrregularlySampledSignal

data at time points series

Event

labeled time point series

Epoch

time periods (time points + durations)

Segment

grouping simultaneously recorded data

Group

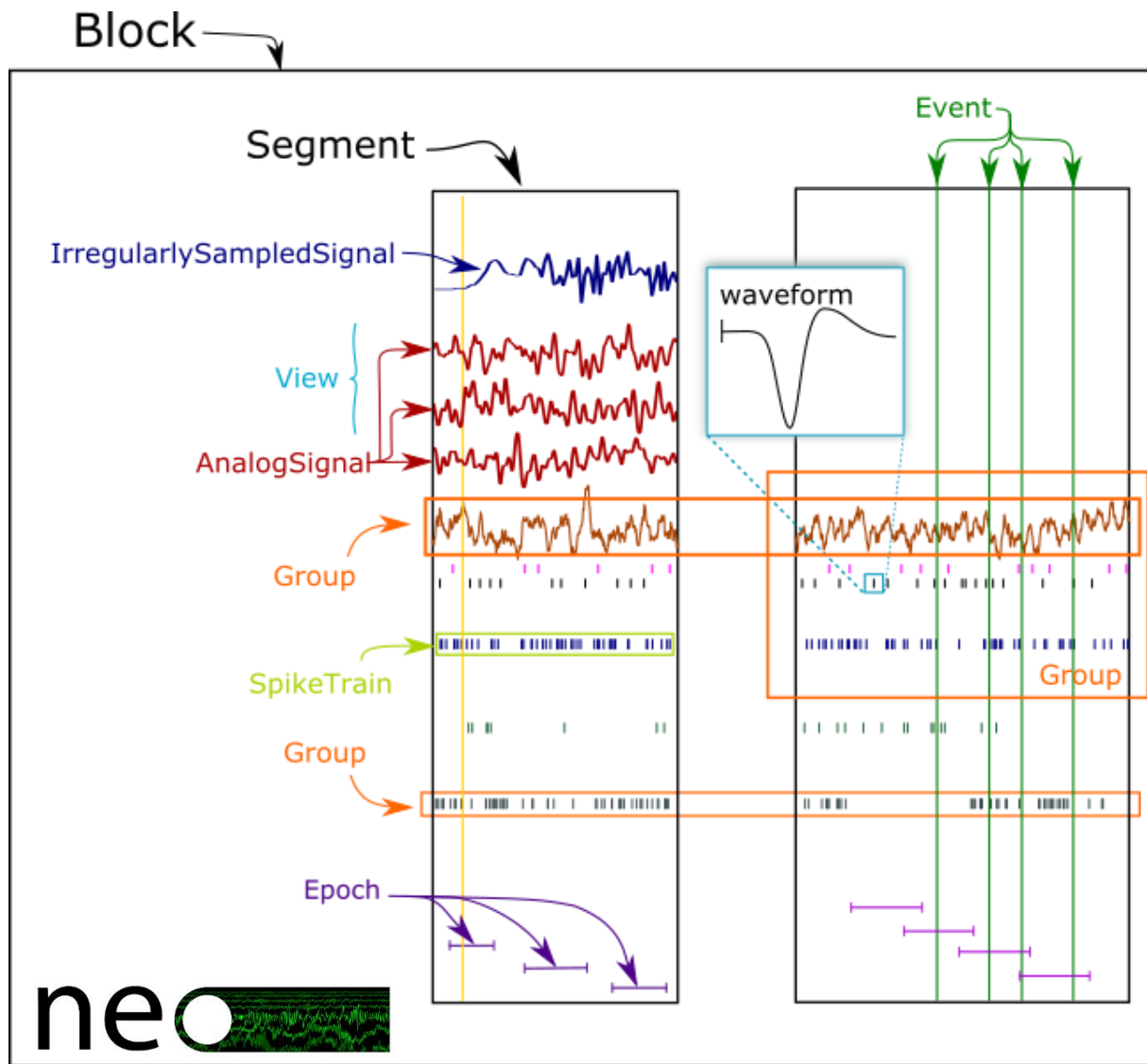
flexible grouping of data objects
within and across segments

ChannelView

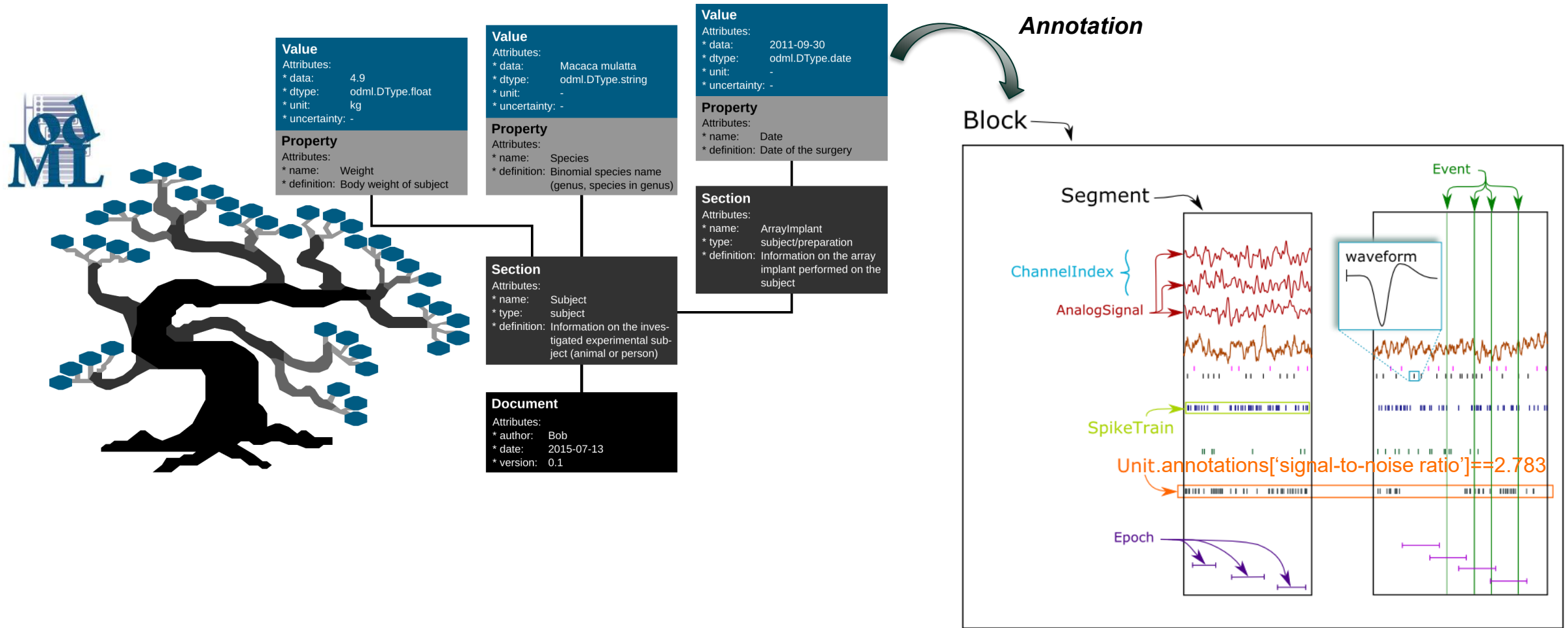
selection of signals within an AnalogSignal

Block

top level grouping of Neo objects

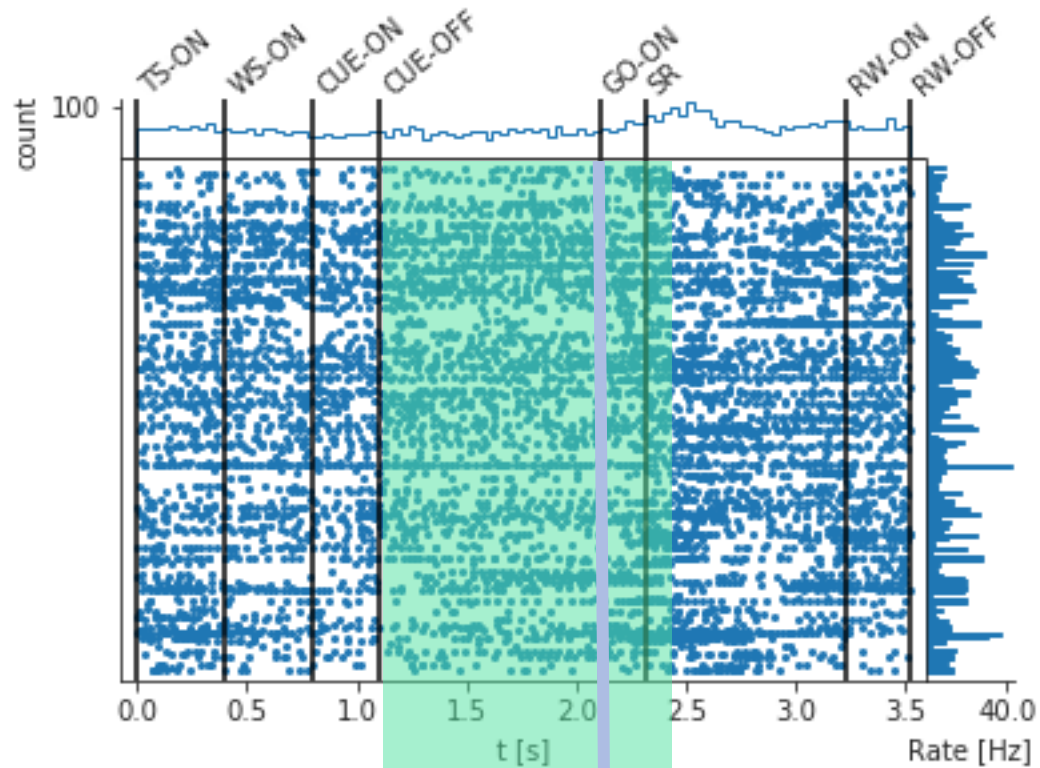


Combining Data and Metadata using Neo



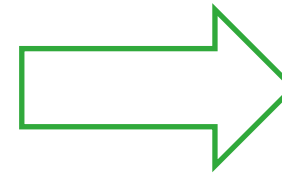
Grewe, Wachtler, Benda (2011) Front. Neuroinf.
Brochier, Zehl, Hao, Duret, Sprenger, Denker, Grün, Riehle (2018) Scientific Data
Zehl, Jaillet, Stoewer, Grewe, Sobolev Wachtler, Brochier, Riehle, Denker, Grün (2016) Front Neuroinf.
Sprenger, Zehl, Pick, Sonntag, Grewe, Wachtler, Grün, Denker (2019) Front. Neuroinf.

Neo for Data Operations



Epoch

Event



Segment Trial 1

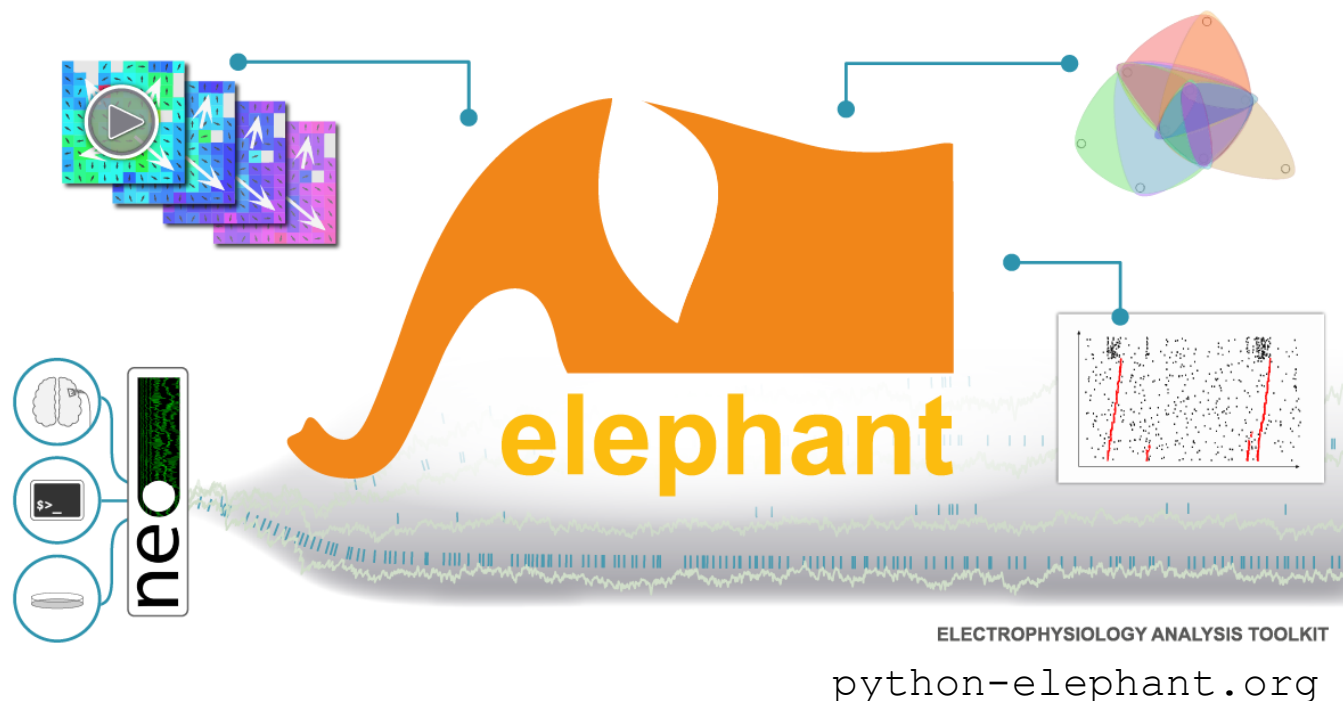
Segment Trial 2

Segment Trial 3

Block

Combining Neo with a Neo-based Tool (Elephant)

Provides generic methods to analyse **brain dynamics** from **experiments** and **simulations** based on the **Neo data model**.



Combining Neo with a Neo-based Tool (Elephant)

Stochastic data

random spike trains
elephant.spike_train_generation
homogenous_poisson_process()

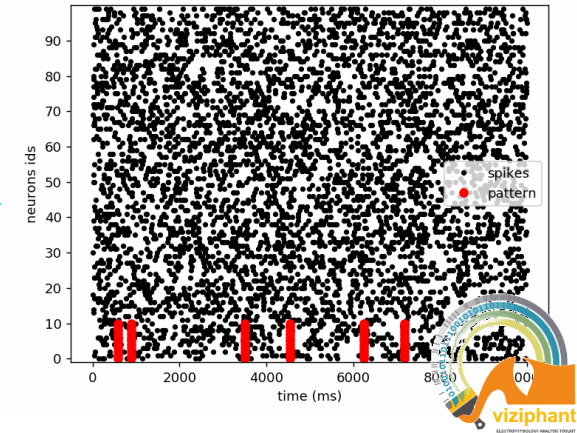
spike trains w/synchr. events
elephant.spike_train_generation
compound_poisson_process()

Experimental data

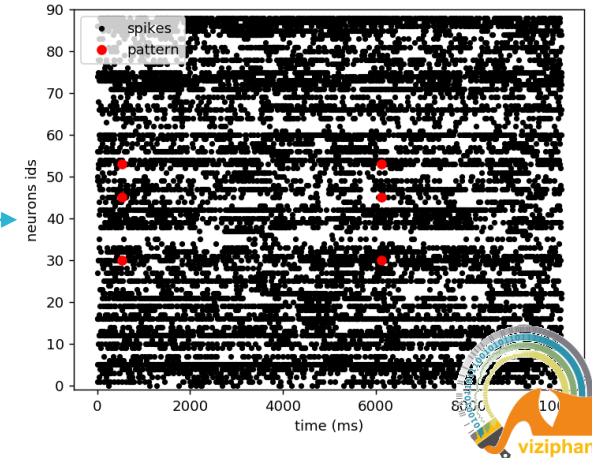
load Utah array recording data
neo.io.blackrockio
BlackrockIO()

select neurons by metadata
neo.Block
filter()

find spatio-temporal patterns
elephant.spade
spade(input: neo.SpikeTrain)



plot the patterns
viziphan.patterns.plot_patterns



Summary

- **Data models** reducing errors, help in tool interoperability, abstract file format decisions, and simplify collaboration.
- **Generic data models (e.g., xarray, nix):** allows richer description of research outputs using given prescriptions for metadata, units, relationship, etc.
- **Domain-Specific data models (e.g., neo):** A library providing data models adapted to represent specific types of research data, with bindings to disk representations (file formats) and applications.